



CS5211 Datasheet

DisplayPort to LVDS Convertor

CapStone Semiconductor Corp.

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1 Introduction

The CS5211 is an eDP to LVDS converter with flexible configuration for the low-cost display system. CS5211 is compliant with eDP 1.2 and supports 1 lane and 2 lanes mode at 1.62Gbps and 2.7Gbps per lane speed. With powerful SerDes technology, CS5211 can recover high-speed serial data with a low bit error rate.

CS5211 LVDS transmitter supports single port and two ports mode. The maximum resolution supported by CS5211 is WUXGA (1920x1200). With 4 configuration pins, CS5211 can support 16 combinations for different panel resolution and LVDS working mode with one EEPROM image. A convenient tool is also provided to edit, generate and update EEPROM image for customized configuration.

With optimized design, the usage of CS5211 is apt to save the BoM cost. The clock source is integrated into this chip, so the external crystal can be saved; with wide range power supporting (1.8 ~ 1.2V for core power), power device on board could be saved; the total power of CS5211 is less than 300mW, so the power supply network design could be simplified. It is easy to integrate the CS5211 to a popular low-cost display system.

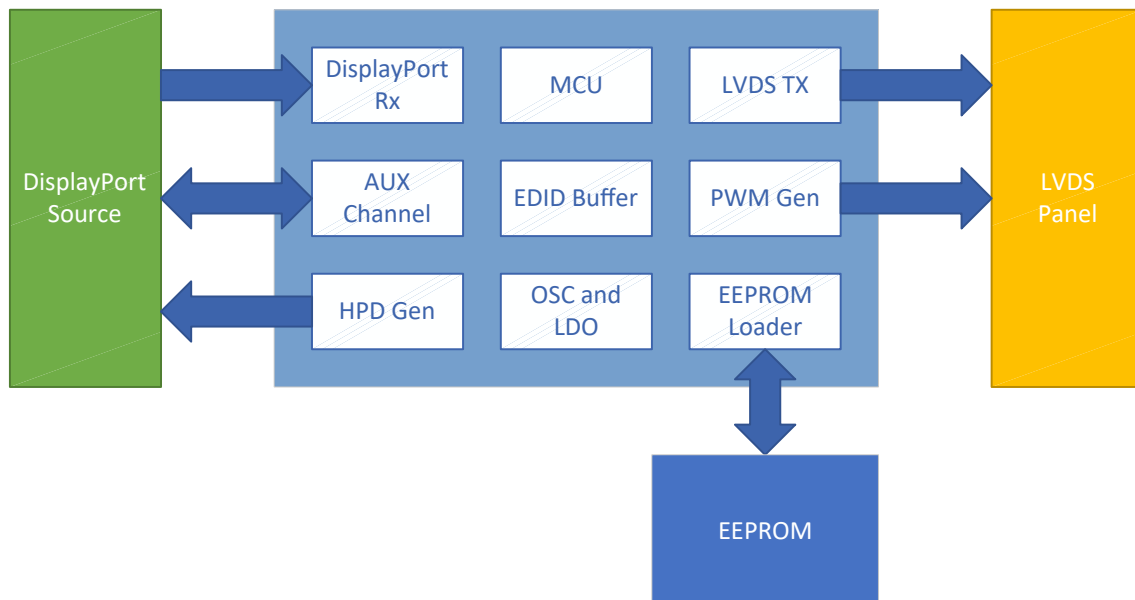


Figure 1-1 CS5211 Block Diagram

2 Features

General

- 2-lane DisplayPort v1.1 compliant receiver
- Support 18-bit Single Port, 18-bit Dual Port, 24-bit Single Port and 24-bit Dual Port LVDS output
- Support both OpenLDI and SPWG bit mapping for LVDS application
- Embedded oscillator and there's no need for the external crystal
- Embedded linear dropout regulator (LDO)
- On chip MCU
- Support panel select by GPIO pins control
- Loads Boot ROM automatically upon power up
- Serial Boot ROM data updated through I2C bus or AUX Channel
- Automatically chip power mode control.
- EMI reduction for both eDP and LVDS
- Wide core power ranges from 1.8V to 1.2V.
- 68-pin QFN package.
- HBM 4KV

DP Receiver

- Compliant with Embedded DisplayPort (eDP) specification 1.2
- Support 1 or 2 main link lane at either 1.62Gbps or 2.7Gbps link rate.
- Input color depth 18/24 bit per pixel in RGB format.
- Support enhanced framing mode.
- Support VESA and CEA timing standards up to 1920x1200 resolution with 60Hz in 24 bit per pixel mode.
- Support dynamic refresh rate switching.
- Support Fast Link Training and Full Link Training.
- Support eDP Authentication: Alternative Scramble Seed Reset and Alternative Framing.
- Support HPD Interrupt.

LVDS Output

- Support 18-bit Single Port, 18-bit Dual Port, 24-bit Single Port and 24-bit Dual Port LVDS output interface.
- Support both OpenLDI and SPWG bit mapping for LVDS application.
- Hold LVDS output when input video is not ready.
- Flexible LVDS output pins swapping.
- Programmable swing/common mode voltage.
- Output slew rate control to reduce EMI.

Video Processing

- Support Gamma Correction.
- Support Dithering and 6 bits + FRC.
- Support Dynamic Backlight Control.

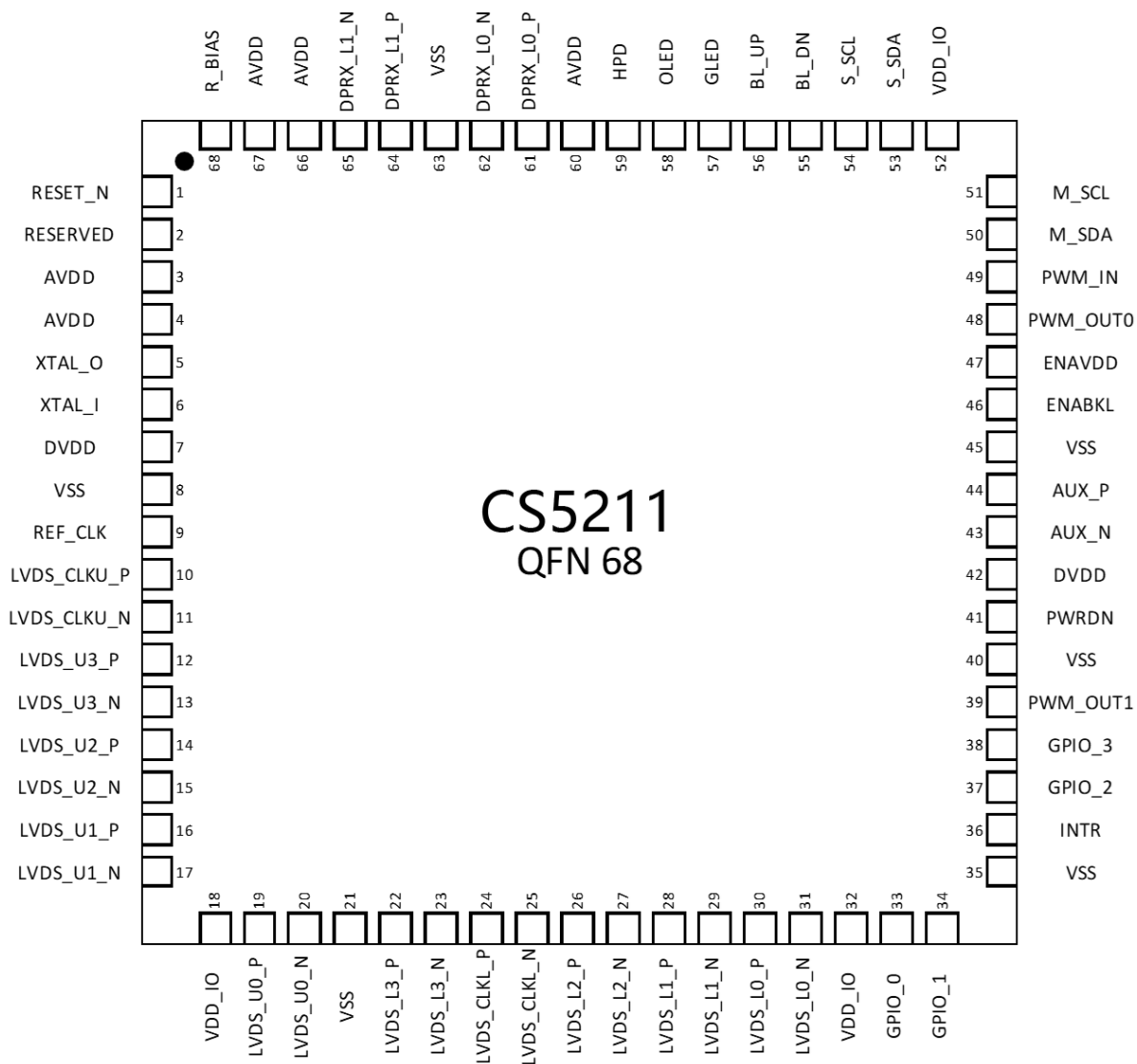
Panel and backlight control

- Programmable LCD panel power sequence
- Supports 2 PWM modes, including backlight luminance level control PWM pin and BLUP/BLDN pin

3 Pin Definition

3.1 Pin Assignments

Figure 3-1 CS5211 Pin Layout



3.2 Pin Description

Table 3-1 CS5211 Pin Definitions

Pin #	Description	Type	Note
1	RESET_N	IN	Reset pin, active low. Internal pull down. When this pin is low, chip is in power down and rest state
2	TEST_EN	IN	Reserved in application. Must left open.
3	AVDD	POWER	Analog Power Supply (1.2 ~1.8V)
4	AVDD	POWER	Analog Power Supply (1.2 ~1.8V)
5	XTAL_O	OUT	Crystal output. Just for backup. By default, integrated clock source will be used.
6	XTAL_I	IN	Crystal input. Just for backup. By default, integrated clock source will be used.
7	DVDD	POWER	Digital Power Supply (1.2 ~1.8V)
8	VSS	GROUND	Power ground
9	REF_CLK	IN	Reference clock input. For debug. By default, integrated clock source will be used.
10	LVDS_CLKU_P	OUT	Positive signal of differential pair for LVDS Up port clock.
11	LVDS_CLKU_N	OUT	Negative signal of differential pair for LVDS Up port clock.
12	LVDS_U3_P	OUT	Positive signal of differential pair for LVDS Up port data lane 3.
13	LVDS_U3_N	OUT	Negative signal of differential pair for LVDS Up port data lane 3.
14	LVDS_U2_P	OUT	Positive signal of differential pair for LVDS Up port data lane 2.
15	LVDS_U2_N	OUT	Negative signal of differential pair for LVDS Up port data lane 2.
16	LVDS_U1_P	OUT	Positive signal of differential pair for LVDS Up port data lane 1.
17	LVDS_U1_N	OUT	Negative signal of differential pair for LVDS Up port data lane 1.
18	VDD_IO	POWER	LVDS driver power supply and IO power supply (3.3V)
19	LVDS_U0_P	OUT	Positive signal of differential pair for LVDS Up port data lane 0.
20	LVDS_U0_N	OUT	Negative signal of differential pair for LVDS Up port data lane 0.
21	VSS	GROUND	Power ground
22	LVDS_L3_P	OUT	Positive signal of differential pair for LVDS Low port data lane 3.
23	LVDS_L3_N	OUT	Negative signal of differential pair for LVDS Low port data lane 3.
24	LVDS_CLKL_P	OUT	Positive signal of differential pair for LVDS Low port clock.

25	LVDS_CLKL_N	OUT	Negative signal of differential pair for LVDS Low port clock.
26	LVDS_L2_P	OUT	Positive signal of differential pair for LVDS Low port data lane 2.
27	LVDS_L2_N	OUT	Negative signal of differential pair for LVDS Low port data lane 2.
28	LVDS_L1_P	OUT	Positive signal of differential pair for LVDS Low port data lane 1.
29	LVDS_L1_N	OUT	Negative signal of differential pair for LVDS Low port data lane 1.
30	LVDS_L0_P	OUT	Positive signal of differential pair for LVDS Low port data lane 0.
31	LVDS_L0_N	OUT	Negative signal of differential pair for LVDS Low port data lane 0.
32	VDD_IO	POWER	LVDS driver power supply and IO power supply (3.3V)
33	GPIO_0	IN	Panel select control signal bit 0.
34	GPIO_1	IN	Panel select control signal bit 1.
35	VSS	GROUND	Power ground
36	INTR	OUT	Programmable interrupt output.
37	GPIO_2	IN	Panel select control signal bit 2.
38	GPIO_3	IN	Panel select control signal bit 3. 4-bits panel select control signal can be combined together to select panel type.
39	PWM_OUT1	OUT	PWM output for backlight brightness control.
40	VSS	GROUND	Power ground
41	PWRDN	IN	Power down control signal. Enter/Exit power down mode when an active low pulse is detected on this pin.
42	DVDD	POWER	Digital Power Supply (1.2 ~1.8V)
43	AUX_N	IN/OUT	Negative signal of differential pair for eDP AUX CH.
44	AUX_P	IN/OUT	Positive signal of differential pair for eDP AUX CH.
45	VSS	GROUND	Power ground
46	ENABKL	OUT	LCD panel back light enable signal.
47	ENAVDD	OUT	LCD panel VCC enable signal.
48	PWM_OUT0	OUT	PWM output for backlight brightness control.
49	PWM_IN	IN	PWM input signal for video source.
50	M_SDA	IN/OUT	I2C data signal for chip boot up and EDID.
51	M_SCL	OUT	I2C clock signal for chip boot up and EDID.
52	VDD_IO	POWER	LVDS driver power supply and IO power supply (3.3V)
53	S_SDA	IN/OUT	I2C data signal for internal register access.
54	S_SCL	IN	I2C clock signal for internal register access.
55	BL_DN	IN	Decrement backlight brightness input.
56	BL_UP	IN	Increment backlight brightness input

57	GLED	OUT	Green LED control signal.
58	OLED	OUT	Orange LED control signal.
59	HPD	OUT	Hot Plug Detect for eDP port.
60	AVDD	POWER	Analog Power Supply (1.2 ~1.8V)
61	DPRX_L0_P	IN	Positive signal of differential for eDP main link lane 0.
62	DPRX_L0_N	IN	Negative signal of differential for eDP main link lane 0.
63	VSS	GROUND	Power ground
64	DPRX_L1_P	IN	Positive signal of differential for eDP main link lane 1.
65	DPRX_L1_N	IN	Negative signal of differential for eDP main link lane 1.
66	AVDD	POWER	Analog Power Supply (1.2 ~1.8V)
67	AVDD	POWER	Analog Power Supply (1.2 ~1.8V)
68	R_BIAS	IN	Band-gap Bias input. Just for backup. By default, left this pin floating.

4 Function Description

4.1 Overview

The CS5211 is a bridge chip that converts the eDP input to LVDS signal. The CS5211's advanced receiver supports Embedded DisplayPort (eDP) 1.1 Specifications. This device will decode high-speed eDP's AC-coupled differential signals in RGB format and drive a TFT panel incorporated with the LVDS interface. A built-in LVDS transmitter, equipped with a flexible OpenLDI/SPWG bit mapping, is capable of driving a single port or a dual port (18/24-bit) LVDS panel. The CS5211's LVDS output can be configured for supporting up to 1920x1200 resolutions at 60 Hz refresh rate. In addition, the CS5211 has display enhancement features including the backlight inverter control, the brightness adjustment, the image dithering algorithm and the EMI reduction mechanism to benefit the system manufactures.

CS5211 is configured as eDP to LVDS converter; The LVDS transmitter supports single port 18-bit, single port 24-bit, dual port 18-bit and dual port 24-bit transmission and supports both OpenLDI and SPWG bit mapping.

4.2 Main Link Receiver

The CS5211's Main link Receiver Functional block is equipped with two eDP Main Link lanes to support high speed, high bandwidth and uni-directional data stream for driving TFT-LCD display up to 1920x1200 resolution. Two types of link rates, 2.7Gbps and 1.62Gbps received through the AC-coupled differential pairs (DP0P/DP0N and DP1P/DP1N pins), is de-serialized into three components (RGB) streams by the CS5211's decoder block.

The CS5211 synchronizes the link rate by using its advanced Clock-Data-Recovery technology, and de-serializes the input stream into parallel data that will be decoded into special symbols based on ANSI8B/10B coding scheme (the channel coding is specified in ANSI X3.230-1994). The decoded symbols are unpacked by CS5211's Main Stream Un-Packer block to restore RGB component video stream and the original video timing information, which will be used by CS5211's LVDS encoding circuits for driving the LCD display.

CS5211 supports eDP authentication through ASSR.

4.3 AUX Channel

The CS5211's AUXP/AUXN pin pair serves as the eDP AUX Channel and supports half-duplex bi-directional AC-coupled differential signal with a 1Mbps data rate for link communication. Upon detection of changing connection status of Hot-Plug pin during system power on, the CS5211 will put its AUX Channel into idle state and wait for a request transaction from the Source Device. When the Source Device needs to initiate a transaction for the Link Service or the Device Service by beginning with the Sync Pattern, which is the Manchester II coding required for AUX Channel communication, will be automatically locked and responded by CS5211's receiver block.

Furthermore, the differential pair allows a Source system to update CS5211's DPCD and retrieve EDID data during the Link Training/Link Maintenance. The Source Device can utilize CS5211's AUXP/AUXN pins for adjusting the panel's luminance level during the normal operation.

4.4 LVDS Panel Selection

Different LVDS Panels may have different EDID and need specific configurations such as power sequence timing. CS5211 supports 16 different LVDS Panel configurations that is selected by GPIO [3:0] pins. CS5211 reads

GPIO[3:0] status after loading its firmware upon power ON and then CS5211 loads LVDS specific configurations. GPIO[3:0] can be controlled by resistors or by other chip's GPIO in real applications.

Method 1 (control by resistors)

GPIO [3:0] can be connected to high/low level by pull-up/pull-down resistors in CS5211 PCB board; CS5211 can get correct LVDS Panel selection value upon power ON or get the reset signal.

Method 2(control by other chip's GPIO)

GPIO[3:0] can be controlled by other chip's GPIO pins in CS5211 application system. The GPIO pins must keep stable values in 100ms after the reset signal is given to CS5211 resetb pin. Otherwise, a reset signal must be given again. It is recommended to reset CS5211 by the controlling chip each time LVDS Panel selection value is changed.

4.5 LVDS-Output Data Mapping

The SPWG and the OpenLDI are two LVDS encoding schemes supported by CS5211 and the selection can be made through programming CS5211 registers. These OpenLDI/SPWG configuration parameters can be stored in the BOOT ROM and loaded into the CS5211 registers during system power-up.

Table 4-1 Signal Mapping for Dual LVDS Channel

	18-bit SPWG / 18-bit OpenLDI	24-BIT SPWG / 24-BIT OPENLDI
LDC[0](1)	R0 / R0	R0 / R2
LDC[0](2)	R1 / R1	R1 / R3
LDC[0](3)	R2 / R2	R2 / R4
LDC[0](4)	R3 / R3	R3 / R5
LDC[0](5)	R4 / R4	R4 / R6
LDC[0](6)	R5 / R5	R5 / R7
LDC[0](7)	G0 / G0	G0 / G2
LDC1	G1 / G1	G1 / G3
LDC[1](2)	G2 / G2	G2 / G4
LDC[1](3)	G3 / G3	G3 / G5
LDC[1](4)	G4 / G4	G4 / G6
LDC[1](5)	G5 / G5	G5 / G7
LDC[1](6)	B0 / B0	B0 / B2
LDC[1](7)	B1 / B1	B1 / B3
LDC[2](1)	B2 / B2	B2 / B4
LDC2	B3 / B3	B3 / B5
LDC[2](3)	B4 / B4	B4 / B6
LDC[2](4)	B5 / B5	B5 / B7
LDC[2](5)	HSYNC / HSYNC	HSYNC / HSYNC
LDC[2](6)	VSYSN / VSYSN	VSYSN / VSYSN

LDC[2](7)	DE / DE	DE / DE
LDC[3](1)		R6 / R0
LDC[3](2)		R7 / R1
LDC3		G6 / G0
LDC[3](4)		G7 / G1
LDC[3](5)		B6 / B0
LDC[3](6)		B7 / B1
LDC[3](7)		RES / RES
LDC[0](1)	Ro0 / Ro0	Ro0 / Ro2
LDC[0](2)	Ro1 / Ro1	Ro1 / Ro3
LDC[0](3)	Ro2 / Ro2	Ro2 / Ro4
LDC[0](4)	Ro3 / Ro3	Ro3 / Ro5
LDC[0](5)	Ro4 / Ro4	Ro4 / Ro6
LDC[0](6)	Ro5 / Ro5	Ro5 / Ro7
LDC[0](7)	Go0 / Go0	Go0 / Ro2
LDC1	Go1 / Go1	Go1 / Ro3
LDC[1](2)	Go2 / Go2	Go2 / Go4
LDC[1](3)	Go3 / Go3	Go3 / Go5
LDC[1](4)	Go4 / Go4	Go4 / Go6
LDC[1](5)	Go5 / Go5	Go5 / Go7
LDC[1](6)	Bo0 / Bo0	Bo0 / Bo2
LDC[1](7)	Bo1 / Bo1	Bo1 / Bo3
LDC[2](1)	Bo2 / Bo2	Bo2 / Bo4
LDC2	Bo3 / Bo3	Bo3 / Bo5
LDC[2](3)	Bo4 / Bo4	Bo4 / Bo6
LDC[2](4)	Bo5 / Bo5	Bo5 / Bo7
LDC[2](5)	HSYNC / HSYNC	HSYNC / HSYNC
LDC[2](6)	VSYNC / VSYNC	VSYNC / VSYNC
LDC[2](7)	DE / DE	DE / DE
LDC[3](1)		Ro6 / Ro0
LDC[3](2)		Ro7 / Ro1
LDC3		Go6 / Ro0
LDC[3](4)		Go7 / Go1
LDC[3](5)		Bo6 / Bo0
LDC[3](6)		Bo7 / Bo1
LDC[3](7)		RES / RES
LDC[4](1)	Re0 / Re0	Re0 / Re2

LDC[4](2)	Re1 / Re1	Re1 / Re3
LDC[4](3)	Re2 / Re2	Re2 / Re4
LDC4	Re3 / Re3	Re3 / Re5
LDC[4](5)	Re4 / Re4	Re4 / Re6
LDC[4](6)	Re5 / Re5	Re5 / Re7
LDC[4](7)	Ge0 / Ge0	Ge0 / Ge2
LDC[5](1)	Ge1 / Ge1	Ge1 / Ge3
LDC[5](2)	Ge2 / Ge2	Ge2 / Ge4
LDC[5](3)	Ge3 / Ge3	Ge3 / Ge5
LDC[5](4)	Ge4 / Ge4	Ge4 / Ge6
LDC5	Ge5 / Ge5	Ge5 / Ge7
LDC[5](6)	Be0 / Be0	Be0 / Be2
LDC[5](7)	Be1 / Be1	Be1 / Be3
LDC[6](1)	Be2 / Be2	Be2 / Be4
LDC[6](2)	Be3 / Be3	Be3 / Be5
LDC[6](3)	Be4 / Be4	Be4 / Be6
LDC[6](4)	Be5 / Be5	Be5 / Be7
LDC[6](5)	HSYNC / LCTLE	HSYNC / LCTLE
LDC6	VSYNC / LCTLF	VSYNC / LCTLF
LDC[6](7)	DE / LA6RL	DE / LA6RL
LDC[7](1)		Re6 / Re0
LDC[7](2)		Re7 / Re1
LDC[7](3)		Ge6 / Re0
LDC[7](4)		Ge7 / Re1
LDC[7](5)		Be6 / Be0
LDC[7](6)		Be7 / Be1
LDC7		RES

4.6 PWM Duty Cycle Modes

The CS5211' has two PWM output pins for panel Backlight Brightness Dimming control, both pins can support duty cycle range from 0% to 100%.

The CS5211 support two PWM control mode: Pass-through mode and TCON PWM output mode. TCON switch two mode according the setting of register DPCD: 0x00721 bit [1:0], this register can be set by DP source.

- Method 1 (Pass-through mode)

In this mode, CS5211 bypass PWM input to PWM output.

- Method 2 (TCON PWM output mode)

In this mode, CS5211 can generate PWM output by itself, the default frequency is 400kHz. In this mode, backlight brightness can be increased or decreased by imposing pulse to BLUP or BLDN pin of CS5211.

CS5211 can dynamically adjust backlight brightness by analyzing brightness level of video data if DYNAMIC_BACKLIGHT_ENABLE is set to 1 by DP Source.

If don't use PWM control, CS5211 can set PWM_OUT pin fixed to high or low by software.

4.7 LVDS Power Sequencing

The CS5211 conforms to the SPWG LVDS panel power sequence requirements. The timing specification shown in Figure 6 is a superset of the requirements in the VESA TV Panels Standard. The control timing parameters of the panel can be set to E2PROM ROM by software and will be programmed into CS5211's Power Control registers when device is powered up.

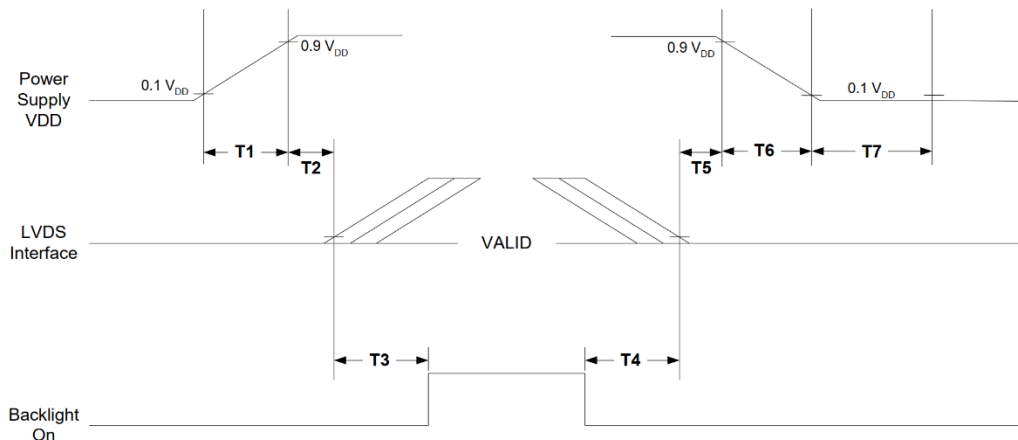


Figure 4-1 Power Supply Sequence

Table 4-2 Power Sequencing

Timing Parameter	Range
T2	0-510 ms
T3	0-32766 ms
T4	0-510 ms
T5	0-32766 ms
T7	0-32766 ms

4.8 LVDS Transmitter

In addition to supporting OpenLDI and SPWG specifications, the CS5211's LVDS transmitter is capable of driving four types of panel electrical interfaces – the 18-bit single channel, the 24-bit single port, the 18-bit dual channel and the 24-bit dual channel. The LVDS output block also supports:

- Flexible Synchronization Control

The HSYNC/VSNC recovered from DisplayPort Main Link can be inverted separately to satisfy the input requirement of a specific LCD Panel. Both Sync signals can also be forced to 0 to support Smart Panels.

- Flexible LVDS Data Output Mapping

The CS5211 has a flexible LVDS output pins mapping for the data and clock to support PCB top/bottom mounting. Please refer to the Register Programming Section how to change the LVDS pin-out mapping.

5 Electrical Specifications

5.1 Absolute Maximum Conditions

Permanent damage may occur if absolute maximum conditions are violated. Refer to Section 5.2 for functional operating limits.

Table 5-1 Absolute Maximum Conditions

Symbol	Parameter	Min	Typ	Max	Unit
DVDD	Digital core supply voltage	-0.3	-	2.16	V
AVDD	Analog core supply voltage	-0.3	-	2.16	V
VDD_IO	Analog I/O supply voltage	-0.3	-	3.96	V
TJ	Junction temperature	-40	—	125	°C
TSTG	Storage temperature ¹	-65	—	150	°C
ESDHB	ESD protection (Human body model)		4		kV

1. Max 260°C can be guaranteed with max 8 sec soldering time.

5.2 Operating Conditions

Table 5-2 Normal Operating Conditions

Symbol	Parameter	Min	Typ	Max	Unit
DVDD	Digital core supply voltage	1.08	1.2-1.8	1.96	V
AVDD	Analog core supply voltage	1.08	1.2-1.8	1.96	V
VDD_IO	Analog I/O supply voltage	2.97	3.3	3.63	V
T _A	Ambient temperature for CS5211AN	0		70	°C
T _A	Ambient temperature for CS5211AN-I	-40		85	°C
Q _{JA}	Package thermal resistance, no air flow	—	39.3	—	°C/W

5.3 Electrical Specification

Table 5-3 DC Electrical Specification

Symbol	Parameter	For 3.3V I/O		
		Min	Typ	Max
V _{il} (V)	Input low voltage	—	—	0.8
V _{ih} (V)	Input high Voltage	2.0	—	—
V _{ol} (V)	Output low voltage	0	—	0.4
V _{oh} (V)	Output high voltage ¹	2.4	—	—
I _{in} (uA)	Input leakage current	-10	—	+10

Symbol	Parameter	For 3.3V I/O		
		Min	Typ	Max
I_{hiz} (uA)	Output tri-state leakage current	-10	—	+10

Table 5-4 AC Electrical Specification

Symbol	Description	Min	Typ	Max	Unit
UI_{High_Rate}	Unit Interval for High Bit Rate (2.7Gbps/lane)		370		ps
UI_{High_Rate}	Unit Interval for High Bit Rate (1.62Gbps/lane)		617		ps
t_{RX-EYE_CONN}	Minimum Receiver Minimum Eye Width at Rx-side connect pins	0.51			UI
t_{RX-EYE_CONN}	Minimum Receiver Minimum Eye Width at Rx package pins	0.47			UI
$T_{RX-EYE-MEDIAN-to-MAX-ITTER_CHIP}$	Maximum time between the jitter median and maximum deviation from the median at Rx package pins			0.265	UI
$L_{RX-SKEWINTER_PAIR}$	Lane-to-Lane Skew at RX package pins			5200	ps
$L_{RX-SKEWINTRA_PAIR}$ High-Bit-Rate	Lane Intra-pair Skew at RX package pins			100	ps
$L_{RX-SKEWINTRA_PAIR}$ _Reduced-Bit- Rate	Lane Intra-pair Skew at RX package pins			300	ps
$F_{RX-TRACKINGBW}$	Jitter Tracking Bandwidth	20			MHz
UI	AUX Unit Interval	0.4	0.5	0.6	us
$T_{AUX-BUS-Pre-charge}$	Number of pre-charge pulse	10		16	

6 Package Specification

Figure 6-1 CS5211 Package Outline (QFN68 Leads 8x8mm²)

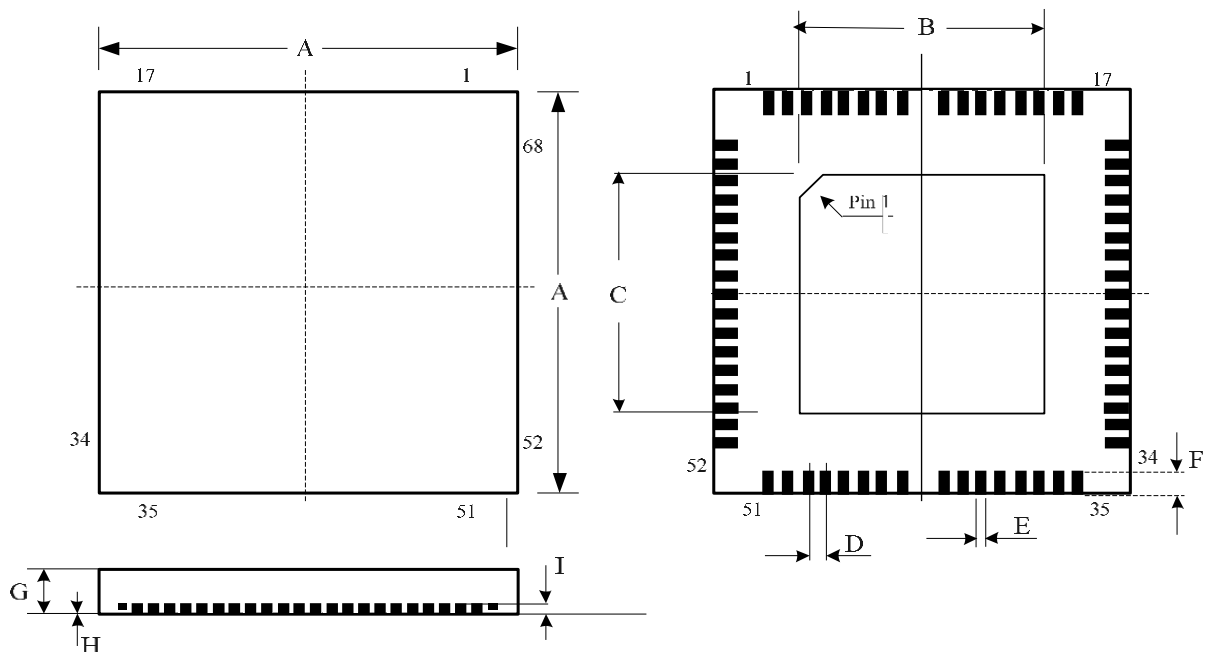


Table 6-1 Package Dimension

Symbol	Dimension in mm			Dimension in inch		
	Min	Normal	Max	Min	Normal	Max
A	7.9	8	8.1	0.311	0.315	0.319
B	4.2	4.3	4.4	0.165	0.169	0.173
C	4.2	4.3	4.4	0.165	0.169	0.173
D		0.40 BSC			0.016BSC	
E	0.15	0.20	0.25	0.006	0.0078	0.0098
F	0.3	0.4	0.5	0.012	0.016	0.020
G	0.8	0.85	0.9	0.031	0.033	0.035
H	0	—	0.05	0	—	0.002
I		0.20 REF			0.008 REF	

7 Ordering Information

The CS5211 can be ordered using the part numbers in Table 7-1. Please consult sales for further details.

Table 7-1 CS5211 Ordering Information

Part No.	Description	Temperature Range	Packing Type
CS5211AN	68 Pin (QFN) Lead-free packages	Industrial : -40 to 85 degree C	Tray

8 Revision History

Table 8-1 Document Revision History

Revision	Date	Changes
Release 1	Sep. 2019	Initial version
Release 2	May 2020	Update
Release 3	June 2020	Update
Release 4	June 2020	Update